DMRN+21: DIGITAL MUSIC RESEARCH NETWORK
ONE-DAY WORKSHOP 2026

KING'S COLLEGE LONDON
FRI 11 DECEMBER 2026

The Archivist

Jason Hockman

Detuned Transmissions (DTND)
School of Digital Arts (SODA), Manchester Metropolitan University, UK, j.hockman@mmu.ac.uk

Historically, fidelity and duration maintained an inverse relation to encode and colour sound. Tape achieved longer durations (lower fidelity) at lower speeds while digital counterparts reduced sample rates. Whereas most modern devices treat fidelity as a single selected parameter or post effect, *The Archivist*,[1] firmware for the Noise Engineering Versio Eurorack platform and agent of Husserl's retention and recollection [1], manipulates sound through a novel performative dimension built atop the fidelity–duration relationship (Fig. 1). `GRAINS` is a microsound [2] processor modelled on the E-mu EII (1984), offering a modeless continuum between reconstruction, pitch and time scaling, cloud generation, Shepard tones, and stochastic sequencing. `TAPE` is the analogue counterpart to `GRAINS`, modelling the Eltro Mark II (c. 1960s) [3], cassettes, and progressive disintegration.

Input audio x is read in overlapping units [4], unit j starting at time t_j from position a_j at pitch ratio p:

$$y(t) = \sum_j g_j(t)\, x\big(a_j + p\,(t - t_j)\big). \tag{1}$$

In `GRAINS`, units are grains spaced d/N apart, with $g_j(t) = g\, w\big((t - t_j)/d\big)$ for gain g and a Hann-to-*exp* window w of duration $d \in [0.05, 0.5]$ s, and $a_j = s\, t_j + \sigma(u)\xi_j + \delta_j$ for speed $s \in [-2, 2]$, Gaussian spray $\sigma(u)\xi_j$ and alignment shift δ_j. The mean overlap (i.e., fill factor [2]) is $N = \min\{N_0 \max(1, |s/p|)\, \Gamma(u),\, N_{\max}\}$, where density sets the base $N_0 \in [0.25, 4]$ with synchronous or stochastic onsets, the cloud factor $\Gamma(u) \geq 1$, with $\Gamma(0) = 1$, grows with ataxy [5] $u \in [0, 1]$, and the device maximum is $N_{\max} = 40$. At $s = p$ and $\sigma = 0$, every grain reads $x(pt)$, so synchronous Hann grains at integer $N \geq 2$ reconstruct; for moderate $s \neq p$ a correlation search (after WSOLA [6]; coarse alignment [7]) sets δ_j for pitch and time scaling, and increased ataxy produces clouds. Above the 1 V/oct range, Shepard tones [8] use 3 octave-spaced, loudness-weighted grain layers gliding at a rate and direction set by s; stochastic sequencing steps p through 12 patterns that mutate, and may revert, on repetition.

In `TAPE`, units are passes of the Mark II's 4 heads, 90° apart on a drum turning against the tape direction; pitch is the head-to-tape speed, $p = r + v$, for tape speed r and head velocity v. Each pass reads from $a_j = r\, t_j + \epsilon_j$, where ϵ_j is placement error, and lasts $T = B/n$ of a bar B, $n = 2^k \leq 32$, so splices, crossfaded by g_j, fall on note values. Disintegration reduces amplitude and high frequencies and adds dropouts, simulating medium degradation over time.

[1] `github.com/jhockman/the-archivist`

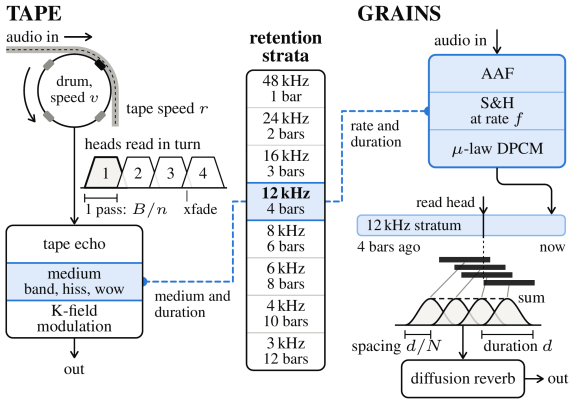


Figure 1: The Archivist's `TAPE` and `GRAINS` modes and retention strata.

Eight retention strata pair rates from 48 kHz down to 3 kHz with spans of 1–12 bars. Below 48 kHz `GRAINS` emulates the EII codec: input is decimated to the stratum rate and coded by 8-bit DPCM with a μ-law companded residual. Grains are selected from the bounding strata with probability set by the fractional position between them. `TAPE` imposes matched band limits, hiss, wow and dropouts at playback. In both modes, deeper strata colour the live signal; upon capture, they mix coarser, longer material with the present. The Archivist thus navigates retention strata to excavate old memories, which find expression through degradation that antiquated media impart.

I. References

[1] E. Husserl, *On the Phenomenology of the Consciousness of Internal Time (1893–1917)*. Kluwer, 1991, trans. J. B. Brough.

[2] C. Roads, *Microsound*. MIT Press, 2001.

[3] W. Carlos, "The Eltro and the voice of HAL," wendycarlos.com, 2008.

[4] B. Truax, "Discovering inner complexity," *Computer Music Journal*, vol. 18, no. 2, pp. 38–48, 1994.

[5] I. Xenakis, *Formalized Music*, rev. ed. Pendragon Press, 1992.

[6] W. Verhelst and M. Roelands, "An overlap-add technique based on waveform similarity (WSOLA) for high quality time-scale modification of speech," in *Proc. IEEE ICASSP*, 1993, pp. 554–557.

[7] É. Gillet, "Clouds (2014) and Beads (2021): Manuals and firmware source," Mutable Instruments.

[8] J.-C. Risset, "Paradoxical sounds," in *Current Directions in Computer Music Research*. MIT Press, 1989, pp. 149–158.